

Paranormal Phenomenon Visual Experience through Binocular Rivalry in Virtual Reality

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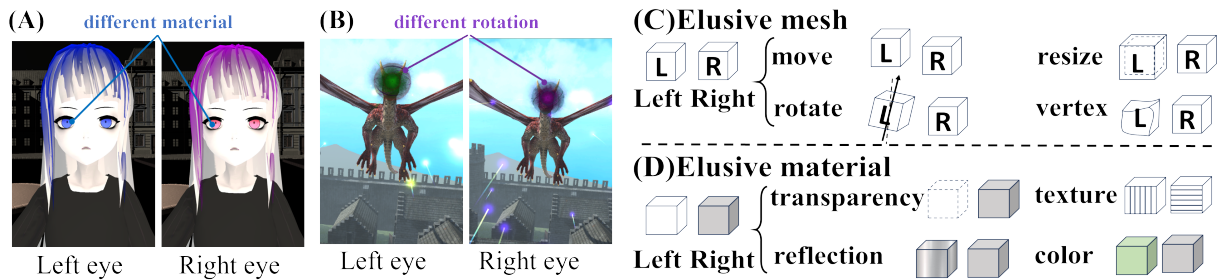


Figure 1: (A) Ghost avatar presentation to each eye in horror demo. (B) Magic effect presentation to each eye in magic demo. (C) Mesh manipulation to induce binocular rivalry. (D) Material manipulation to induce binocular rivalry.

ABSTRACT

Visual games and media design frequently depict paranormal visual effects like ghosts and magic. However, current methods do not achieve realistic paranormal experiences. This study proposed new visual presentation methods that utilize binocular rivalry to provide a more realistic visual experience of paranormal phenomena and enhance user satisfaction with magical effects and horror effects.

CCS CONCEPTS

• **Human-centered computing** → **Virtual reality; Mixed / augmented reality.**

KEYWORDS

Virtual Reality, Visual Perception, Binocular Rivalry, HCI

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1 INTRODUCTION

Imaginary paranormal phenomena, which exist in dimensions distinct from our world, have been deeply rooted in human culture and continue to impact our lives. Ghosts and magic, as examples of paranormal phenomena, are prevalent themes in video games and media design. In textual descriptions, paranormal phenomena are often depicted as “invisible” or possessing “visual elusiveness,” conveying the sensation of perceiving a presence without discerning its features [Janik 2019]. On the other hand, current methods for creating paranormal visual effects primarily involve using transparent materials or shaking effect [Perron 2009]. However, these methods only symbolically represent paranormal phenomena, producing visually perceptible sensations but lacking cognitive engagement. Since paranormal experiences resemble a sixth sense, relying on visual effects cannot deliver an authentic paranormal experience.

To effectively represent the elusive visual sensations attributed to a sixth sense, we focus on a phenomenon known as binocular rivalry, which has the potential to produce paranormal sensations both visually and cognitively. Binocular rivalry occurs when dissimilar images are presented to each eye. During this phenomenon, the two images do not blend into a coherent view. Instead, they compete for perceptual dominance: one image can dominate conscious awareness for several seconds, then be supplanted by the previously suppressed rival image [Breese 1909]. In binocular rivalry, our awareness of images can be influenced by attention [Paffen and Alais 2011]. However, it is impossible to quickly switch between awareness images or to maintain dominance of one image consistently. As a result, the fluctuation between controllable and

uncontrollable awareness creates a sense of elusiveness in cognitive perception, as if something is visible yet remains indiscernible. Additionally, since binocular rivalry is rarely observed under normal conditions, it offers a unique visual experience [Lack 1974]. Consequently, we believe the sensation of being only partially visible, caused by the lack of complete control, combined with this unique visual experience, is suitable for paranormal visual presentations.

2 METHOD

We explored two design principles concerning the fundamental components of the object: mesh and material, in order to create the sensation of paranormal phenomena.

2.1 Elusive mesh

To create a sense of elusive mesh, we focus on the rivalry between two distinct meshes of images presented to each eye, as shown in Figure 1 (C). However, we face the challenge that overly similar meshes do not induce binocular rivalry. In contrast, extremely different meshes make it difficult for the brain to reconcile them as representations of the same object. Therefore, we propose the following design principles: 1) Manipulate the target mesh by moving, rotating, resizing, and altering its shape to induce binocular rivalry. 2) Different mesh manipulations, as previously mentioned, have varying thresholds. Designers need to test these threshold ranges to ensure consistent cognition of the same object.

2.2 Elusive material

This section focuses on another fundamental aspect of objects: material. We have established flowing design principles for creating an elusive material experience: 1) As shown in Figure 1 (D), the material presented to the left and right eyes should differ in color, texture, etc. 2) Considering that the dominant eye can influence the status of binocular rivalry, designers should take into account the user's dominant eye and, if necessary, dynamically switch the materials shown to each eye to manage binocular rivalry states.

3 PRELIMINARY USER STUDY

We conducted two demonstrations: a Horror Demo and a Magic Demo, to carry out a preliminary user study that verifies the effects of the proposed methods on the paranormal experience. Each demo comprises a binocular rivalry version and a normal version. The binocular rivalry version, as shown in Figure 1 (A) and (B), adopts our proposed method. The normal version uses current methods, where the same image is seen in both eyes. In this study, we set the view of the right eye in Figure 1 (A) and (B) to be displayed in both eyes for the normal version.

3.1 Horror Demo

As illustrated in Figure 1(A), in the binocular rivalry version, avatars were displayed with the only difference in the color of the eye region and the transparent meshes to induce rivalry. During avatar movement, blue transparent meshes were displayed as trials for only one eye to induce binocular rivalry between the blue transparent meshes and the background environment.

3.2 Magic Demo

Similarly to the design techniques detailed in Section 3.1, we adjusted the materials of the magic effect in the left and right eyes.

Following the methods mentioned in Section 2, this adjustment involved using different colors and altering the mesh by applying different rotations, as illustrated in Figure 1(B).

3.3 Feedback of demonstration

Eight volunteers participated in our two demonstrations, each with two conditions: a normal version and a binocular rivalry version. After each demonstration, participants were asked to rate their feelings of 'fear,' 'sense of wonder,' and 'paranormal sensation' on a 7-point scale. Figure 2(A) shows the responses regarding paranormal sensation and fear sensation in the horror demo, while Figure 2(B) shows the responses regarding paranormal sensation and sense of wonder in the magic demo. The feedback responses provided the following comments: "The visual experience brought by binocular rivalry is unique and highly suitable for paranormal phenomena." and "The magic effects utilizing binocular rivalry gave me a unique and amazing experience that I had never felt before." Additionally, all participants mentioned that our proposed methods are suitable for paranormal phenomena. Seven out of eight participants preferred using binocular rivalry methods in horror games, and six out of eight preferred using binocular rivalry methods in magic effect games. These results suggest that the proposed binocular rivalry presentation methods offer users a more authentic paranormal experience compared to existing methods.

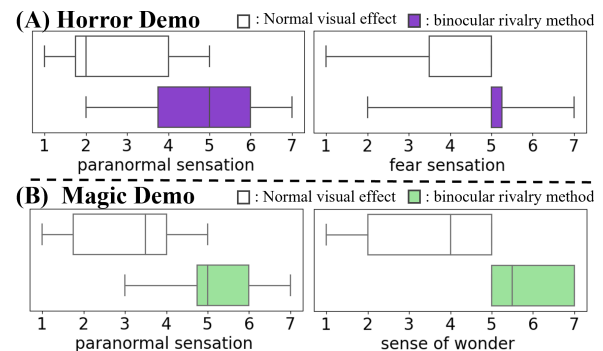


Figure 2: (A) Response results in Horror Demo. (B) Response results in Magic Demo.

4 USER EXPERIENCE

Our proposed methods aim to create more realistic paranormal sensations by binocular rivalry. Our preliminary study also suggests that our methods can provide users with a unique experience, making them useful for providing realistic VR/AR paranormal experiences. Furthermore, our concept of presenting different images to each eye is not only for paranormal visual presentations but also has the potential to represent other visual effects.

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